

Grey leaf spot of maize

Cercospora zeae-maydis, *Cercospora zeina*

	Prevention	Monitoring	Direct Control
 Maize growing in Zambia showing symptoms of grey leaf spot (Noah Phiri, CABI)  Spots join together resulting in long grey streaks and a blight of the leaf (Noah Phiri, CABI)	• Use tolerant varieties if available. • Till the land to bury and kill the disease. • Plant crops away from, and upwind of, fields with the disease as it can travel by wind. • If direct seeding (no-tillage or zero tillage), grow legume cover crops before planting, e.g. lablab, velvet bean or sunhemp. Clear the legume just before planting the maize by making furrows without cultivating the land. • At first weeding, plant a short-term food legume cover crop such as beans, peas, mung bean or soybean as an intercrop to cover the soil. • Avoid overhead irrigation, if possible, as wet plants can encourage disease development. • Rotate crop at least every 1-2 years. The disease only infects maize, so other crops in the rotation will not become infected. • Use fertilizer moderately. High levels of nitrogen in the soil has been associated with increased disease severity.	• Look out for: • Pinpoint spots surrounded by yellow halos on lower leaves (best seen when held to the light). • Expansion of the spots parallel to the leaf veins, sometimes light brown to grey, rectangular, up to 70 mm long and 4 mm wide. • Spots may join together to form a blight and may turn silvery-grey. The blight can cover the whole leaf. • Spots can also sometimes be found on stalks and leaf husks. • Rotting of the sheath and stalk causing the plant to fall over. • Regularly check crop, especially when full canopy has developed, as the resulting high relative humidity in the canopy encourages disease development. • Favourable conditions for disease development include high humidity and temperatures between 22 and 30°C.	• After harvest, plough in or collect and burn crop debris. This is very important as the fungus can survive on the leaves and stalks from one season to another.

Note: Pesticides may be available to control this pest. Please check with the Ministry of Agriculture in your country to find out which pesticides are registered in your country and the local restrictions for their use.